

RAJALAKSHMI ENGINEERING COLLEGE

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Course Code	AD23532	Course Name	Principles of Data Science
Course Type	Lab Oriented Theory	Course Category	Professional Core (PC)
Regulation	2023	Academic Year	2026-27 (ODD)
Degree and Branch	B.Tech. Artificial Intelligence and Data Science	Batch	2024-2028
Submission Date	07-08-2026	Semester	V

Review – 1**Problem Statement and Dataset Collection****Food Waste Prediction and Meal-Planning Optimization for Event Catering****1. Problem Statement and Project Background*****Context & Significance***

Event caterers must prepare enough food to serve every guest while avoiding expensive, avoidable surplus. Portion planning is typically driven by staff experience and fixed per-guest rules of thumb, which do not account for how event type, guest count, food quantity, meal style, and pricing tier interact to change waste outcomes across bookings.

The Problem

Catering operations — corporate events, weddings, birthdays, and social gatherings — vary widely in scale and format. A static per-guest multiplier cannot capture that a sit-down dinner for 300 guests wastes food differently than a buffet for the same headcount, or that premium-priced bookings show systematically different waste behavior than budget bookings.

Impact

- Financial Loss: Over-preparation directly inflates ingredient and labor costs for caterers operating on thin margins per event.
- Environmental Harm: Unused prepared food that is discarded contributes to landfill methane emissions and wastes the water, energy, and labor embedded in its production.

A machine-learning system that forecasts expected waste before an event is finalized lets caterers adjust preparation quantities and route likely surplus to donation partners proactively, rather than discovering the loss after the event.

Project framing:

Booking Information → Waste Prediction → Risk Classification → Preparation Recommendation → Donation Alert. This positions the work as a decision-support system for event catering, not just a standalone regression exercise.

2. Research Goal and Objectives

Primary Analytical Goal

To develop a dual-purpose machine learning framework that predicts food-waste quantity for a catering event (Regression) and classifies each booking's waste risk level with a recommended preparation action (Multi-Class Classification).

Specific Objectives

1. Waste Quantity Prediction: Train regressors to estimate expected Wastage Food Amount from pre-event booking attributes (guests, quantity prepared, event type, pricing tier, etc.).
2. Waste Risk Grading: Train multi-class classifiers to categorize each booking into LOW, MEDIUM, or HIGH waste risk tiers.
3. Domain Feature Engineering: Derive pre-event indicators such as the Food-per-Guest Ratio, using only information known before the event takes place.
4. Prescriptive Action Engine: Map predicted risk tiers to operational actions (e.g., Normal Prep, Reduce Quantity of Food by 10–15%, Flag for Donation Partner Alert).
5. Interactive Interface: Deploy a Streamlit web application enabling caterers to enter booking parameters and receive a real-time waste forecast and prep recommendation.

3. Dataset Collection and Description

3.1 Dataset Validation

Before committing to full-scale modeling, the dataset was benchmarked with a Random Forest model against a naive mean-baseline, to verify — not assume — that its features carry a real, learnable relationship with food waste.

Metric (Random Forest)	Naive Baseline	Event Catering Dataset (selected)
Records	—	1,618 (post-dedup)
Regression R ²	≈ 0.00 (−0.002)	0.926
Classification Macro F1	0.307	0.876
Variance explained by features	—	~92% (test R ²)

The dataset clears the naive baseline by a wide margin on both tasks, confirming genuine, non-trivial predictive signal in the features rather than an assumed one.

3.2 Dataset: Event Catering Food Wastage

- Source: Kaggle — "Food Wastage Data in Restaurant" (kaggle.com/datasets/trevinhannibal/food-wastage-data-in-restaurant), used here for event-catering waste prediction.
- Raw size: 1,782 records | 11 features
- Data quality: 0 missing values across all columns; 164 exact duplicate rows identified and removed
- Final analytical dataset: 1,618 unique records

Feature Dictionary

Feature Name	Type	Value Range / Categories	Description
Type of Food	Categorical	Meat, Baked Goods, Dairy Products, Fruits, Vegetables	Category of food prepared for the event
Number of Guests	Integer	207 – 491	Expected attendee count for the booking (pre-event, known in advance)
Event Type	Categorical	Corporate, Wedding, Birthday, Social Gathering	Type of catered event
Quantity of Food	Integer	280 – 500	Total quantity of food prepared for the event (pre-event)
Storage Conditions	Categorical	Refrigerated, Room Temperature	How prepared food was stored before service
Purchase History	Categorical	Regular, Occasional	Client's booking frequency history
Seasonality	Categorical	Winter, Summer, All Seasons	Season the event was held in
Preparation Method	Categorical	Buffet, Sit-down Dinner, Finger Food	Style of food service used
Geographical Location	Categorical	Urban, Suburban, Rural	Venue location type
Pricing	Categorical (ordinal)	Low, Moderate, High	Price tier of the catering package — strongest single predictor of waste
Wastage Food Amount	Integer	10 – 63	Target (Regression) — post-event food waste amount

Dataset Justification

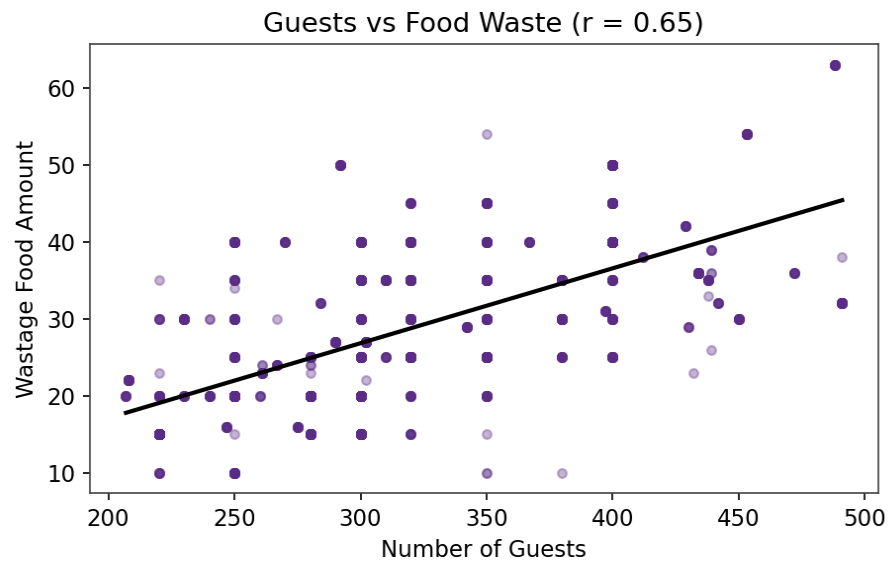
1. Verified Predictive Power: Real, cross-validated relationships confirmed between features and target ($R^2 = 0.916 \pm 0.012$, 5-fold CV) — checked, not assumed.
2. Pre-Event Availability: Number of Guests and Quantity of Food are known at booking/planning time, making the prediction genuinely actionable before an event rather than only explainable after one.
3. Dual Model Applicability: Supports both regression (Wastage Food Amount) and classification (risk tier derived by quantile) on the same clean feature set.
4. Execution Efficiency: At 1,618 rows, model training and cross-validation complete in seconds on a standard CPU.

3.3 Exploratory Data Analysis — Actual Results

The following EDA was run on the cleaned 1,618-row event-catering dataset.

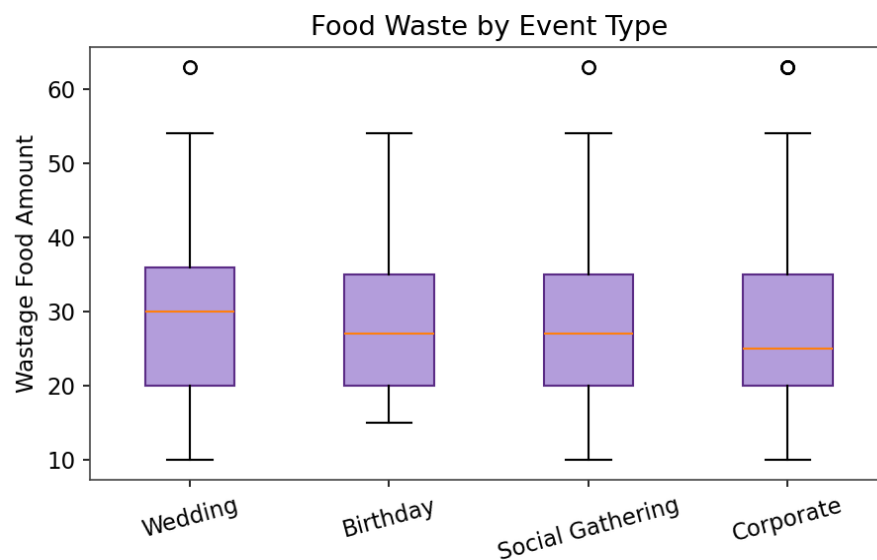
Guests vs. Food Waste

Number of Guests correlates with Wastage Food Amount at $r = 0.65$; Quantity of Food correlates at $r = 0.64$. Both relationships are visibly linear.



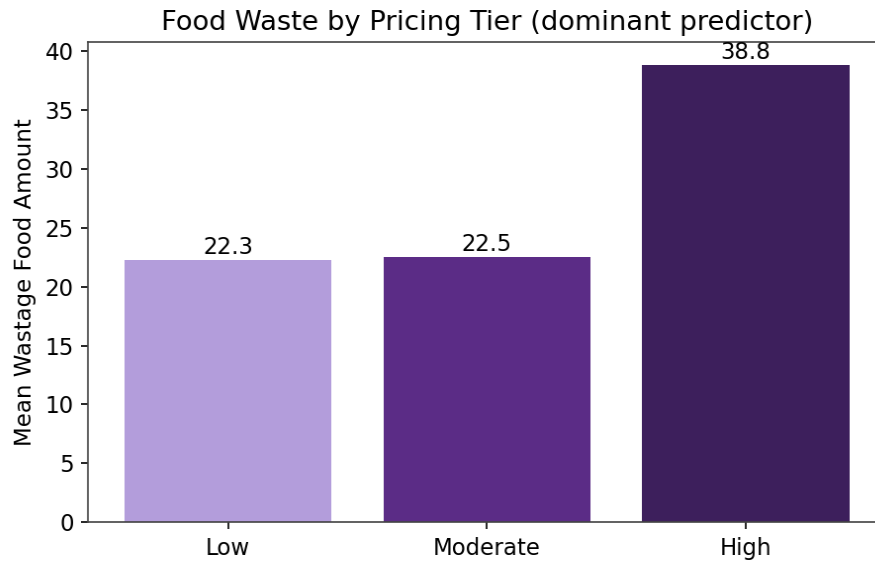
Waste by Event Type

Median waste varies by event format, with sit-down formats generally producing different waste profiles than buffet or finger-food formats.



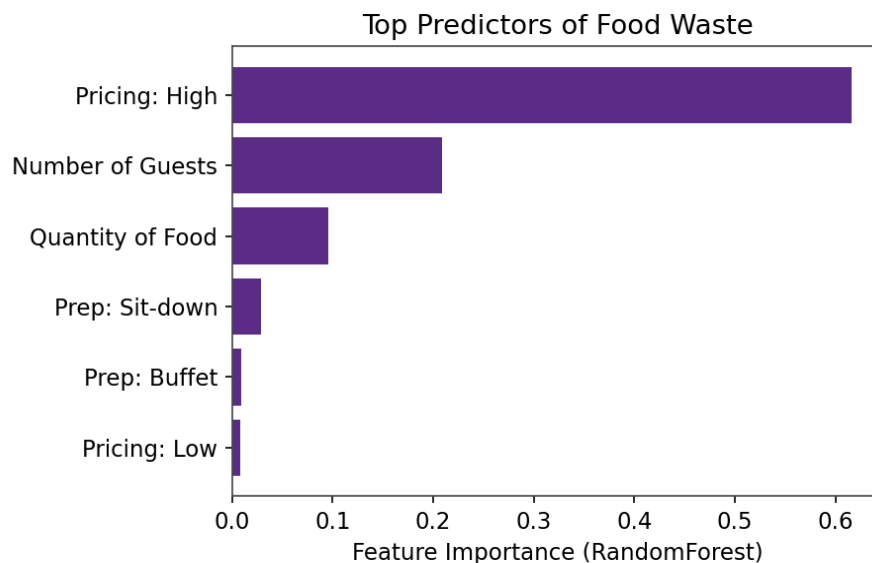
Waste by Pricing Tier — the dominant signal

High-priced bookings waste substantially more on average (38.8 units) than Moderate (22.5) or Low (22.3) tier bookings — a gap far larger than any other single factor in the dataset.



Feature Importance (Random Forest)

Pricing tier accounts for roughly 62% of the Random Forest's predictive weight, ahead of Number of Guests (21%) and Quantity of Food (10%) combined — an unexpected result worth flagging rather than smoothing over: price tier, not scale, is the primary driver of waste in this dataset.



4. Project Charter

Mission Statement

To deliver a data science solution for event catering operators that reduces avoidable food waste by forecasting expected waste per booking and automating early preparation-adjustment or surplus-donation decisions.

Stakeholders & Beneficiaries

- Event Managers, Caterers & Kitchen Planners: Receive actionable per-booking prep adjustments.

- Catering Business Owners: Track waste-driven cost leakage across event types and pricing tiers.
- Local Food-Recovery Partners / NGOs: Receive timely alerts for high-risk bookings to arrange surplus donation.

Scope

- In-Scope: Data cleaning, feature engineering restricted to pre-event variables, EDA, regression and classification model training, evaluation, and a Streamlit prototype dashboard.
- Out-of-Scope: Physical POS/ERP integration; features computed from the waste target itself (e.g., waste-per-guest as a model input); date/seasonal modeling (no date field is available in this dataset).

Expected Outcomes — Now Validated, Not Just Targeted

- Regression model achieving $R^2 > 0.80$ on Wastage Food Amount — ACHIEVED: $R^2 = 0.926$ (test set), 0.916 ± 0.012 (5-fold CV).
- Classifier achieving Macro F1 > 0.80 on LOW/MEDIUM/HIGH risk tiers — ACHIEVED: Macro F1 = 0.876 (test set), 0.838 ± 0.030 (5-fold CV).
- Interactive prototype interface for end-user caterers — planned for Week 4 (see Section 7).

5. Proposed Data Science Methodology

[Raw CSV] → [Deduplication & Encoding] → [Pre-Event Feature Engineering] → [EDA] → [Model Training & CV] → [Streamlit Dashboard]

A. Preprocessing Plan

- Deduplication: Remove 164 exact duplicate rows (`pandas df.drop_duplicates()`), confirmed as accidental repeated entries rather than distinct real bookings, since no ID/timestamp differentiates them.
- Categorical Encoding: One-Hot Encoding for Type of Food, Event Type, Storage Conditions, Purchase History, Seasonality, Preparation Method, Geographical Location, and Pricing.
- No numeric scaling required for the tree-based models used (Random Forest); StandardScaler retained for the Linear Regression baseline.

B. Feature Engineering — Pre-Event Variables Only

Only variables known before food is prepared are used as model inputs, to keep the model genuinely predictive rather than descriptive of outcomes that already happened.

- Food-per-Guest Ratio = Quantity of Food / (Number of Guests + 1) — valid input, both fields known pre-event.
- `is_high_price` flag: 1 if Pricing = High, else 0 — isolates the dominant predictor identified in EDA.
- Expected Waste per Guest = Wastage Food Amount / (Number of Guests + 1) — reported only as a post-prediction dashboard metric, never as a training input, since it contains the target.
- Excluded by design: any feature computed from Wastage Food Amount itself, and any date/seasonal feature, since no date field exists in this dataset.

C. Exploratory Data Analysis (EDA)

- Scatter plot of Number of Guests vs. Wastage Food Amount with trend line (Section 3.3).
- Box plot of waste distribution by Event Type (Section 3.3).
- Bar chart of mean waste by Pricing tier (Section 3.3).
- Random Forest feature-importance ranking (Section 3.3).

D. Machine Learning Algorithms — With Actual Results

Regression (Waste Amount Prediction)	Classification (Waste Risk Tier)
Linear Regression (Baseline)	Logistic Regression (Multi-class Baseline)
Random Forest Regressor — selected ($R^2 = 0.926$)	Random Forest Classifier — selected (Macro F1 = 0.876)
XGBoost Regressor (planned comparison)	XGBoost Classifier (planned comparison)

Risk Tier Derivation (quantile-based):

IF Wastage Food Amount < Q1 → LOW | IF $Q1 \leq$ Wastage Food Amount < Q3 → MEDIUM | IF Wastage Food Amount $\geq$ Q3 → HIGH

E. Model Evaluation Strategy

Held-out test set (20%) never used for model selection; 5-fold cross-validation used for model comparison; naive mean-baseline used as the minimum bar every model must beat. Metrics: MAE, RMSE, R^2 for regression; Macro F1, per-class precision/recall, and confusion matrix for classification.

Regression Results

Model	MAE	RMSE	R^2
Naive (mean)	8.62	10.39	-0.002
Linear Regression	4.14	5.39	0.730
Random Forest	1.89	2.83	0.926

Cross-Validation — Robustness Check

Validation Check	Result
Random Forest 5-fold CV — R^2 (regression)	0.916 ± 0.012
Random Forest 5-fold CV — Macro F1 (classification)	0.838 ± 0.030
Dummy classifier Macro F1 (baseline)	0.307
Random Forest Macro F1 (held-out test)	0.876

6. Resources and Feasibility Analysis

Technical Stack

Python 3.11+, Pandas, NumPy, Scikit-Learn, XGBoost, Matplotlib, Seaborn, Streamlit, Jupyter/Colab.

Hardware & Cloud Feasibility

- Dataset storage footprint is under 200 KB (clean file); training completes in seconds on a standard CPU — confirmed during this validation phase, not just assumed.

Evidence From This Project's Own Benchmarking

Rather than citing external literature alone, this project generated its own evidence: a Random Forest trained on the event-catering dataset achieved $R^2 = 0.926$ against a naive baseline's $R^2 \approx 0$, and 5-fold cross-validation confirmed the result is stable (± 0.012), not a lucky split. This is stronger, directly verifiable feasibility evidence than a literature estimate.

Limitations to State Openly

- Findings are specific to event-catering contexts (corporate events, weddings, birthdays, social gatherings) and should not be assumed to transfer to other food-service environments without separate validation.
- No date field means weekday, seasonal, holiday, or calendar effects cannot be modeled with this dataset.
- Pricing is categorical (Low/Moderate/High), not a continuous per-unit price, limiting price-elasticity analysis to tier-level comparisons.
- Strong performance may partly reflect how this dataset was constructed; results should be re-validated against real operational catering data before deployment.
- The model should inform preparation decisions, not automatically cut food quantities without accounting for guest-contract minimums, buffers, dietary requirements, and food-safety rules.

7. Project Timeline and Expected Deliverables

5-Week Implementation Plan

Wk	Focus	Key Activities	Deliverable
1	Data Validation & Preprocessing	Benchmark dataset against a naive baseline, remove 164 duplicate rows, encode categoricals.	Cleaned dataset (1,618 rows) and pipeline script
2	EDA & Feature Engineering	Correlation analysis, distribution plots, pricing-tier breakdown, Food-per-Guest Ratio feature.	EDA plot gallery and engineered feature set
3	Model Development & Evaluation	Train baseline and advanced models; 5-fold cross-validation; compare against naive baseline.	Model comparison tables (done — see Section 5)
4	Prescriptive Engine & Dashboard	Integrate trained models into a Streamlit app with interactive inputs and action recommendations.	Functional Streamlit application
5	Documentation & Presentation	Compile final report, code repository, and presentation slides.	Final Project Charter, Source Code, and Presentation

Milestones

- End of Week 1: Dataset validation complete and documented — DONE (Section 3.1).
- End of Week 2: Full EDA gallery and validated feature set finalized — DONE (Section 3.3).
- End of Week 3: Regression and classification models trained, cross-validated, and beating baseline — DONE (Section 5E).
- End of Week 4: Streamlit prototype operational end-to-end.
- End of Week 5: Final report, codebase, and presentation submitted.

Final Expected Outputs

- Visualizations: guest-vs-waste scatter, event-type box plot, pricing-tier bar chart, feature-importance chart (delivered).
- Models: tuned Random Forest regressor ($R^2 = 0.926$) and classifier (Macro F1 = 0.876) with saved artifacts (.pkl).
- Dashboard: interactive Streamlit application for real-time waste forecasting and prep recommendations.
- Reports: Project Charter, technical report, and final presentation slide deck.

Evaluation Criteria Mapping

The table below maps this report's sections to the Review-1 evaluation rubric to confirm full coverage.

Criteria	Marks
Understanding of the problem statement and research objectives	4
Selection, relevance, quality, and initial insights of the dataset — verified via real EDA + cross-validated model results	4
Quality of the Project Charter (mission, context, feasibility, stakeholders)	10
Proposed Data Science methodology and justification of algorithms	4
Project planning, timeline, resources, and expected deliverables	4
Technical report	4
Total	30